ThoughtFlow-FP8: A Mixed Gate for Phase-Aware KV Retention

Anonymous authors
Paper under double-blind review

Abstract

ThoughtFlow-FP8 tests whether explicit anchor/phase retention plus byte-aware FP8-style budgeting can preserve useful reasoning context better than matched-budget KV-retention proxies. Synthetic traces support the intuition that phase markers and anchors can be protected, but real retained-context quality weakens the branch. On 24 distilgpt2 traces at a 0.20 keep fraction, a bounded successor policy, ThoughtFlow-saliency-recent, improves over the original phase-only policy and beats LongFlow-like and ThinKV-like proxies, but narrowly loses to an R-KV-like retained-prefix proxy on continuation NLL (3.434 vs 3.419). A held-out sweep over recency and saliency weights then finds a matched-budget ThoughtFlow-family policy that ties R-KV-like on held-out traces (3.480 vs 3.482 NLL; positive margin +0.001). We add real hidden/KV telemetry from distilgpt2 attention, final-hidden norms, key norms, value norms, and combined KV norms. It is diagnostic but still mixed: ThoughtFlow beats the strongest real-saliency proxy on phase recall, while math-state recall has an uncertain paired margin and the local LongFlow-like importance proxy ties it. We therefore present this as a mixed workshop gate, not a positive method. The next useful step is actual cache-dropping or sparse-KV quality validation for a train-fixed successor policy.

1 Policy Family

The original policy protects anchors and phase labels. The first successor adds a small recent-token reserve. The second successor protects anchors, reserves half the retained budget for recent tokens, and lets phase markers compete with math-state and high-importance reasoning tokens under a saliency bonus. This tests whether the failure was caused by over-preserving interpretable markers that do not matter for immediate continuation likelihood.

Policy	Keep rate	NLL	Δ NLL vs full	PPL
full context	1.000	2.101	0.000	9.7
R-KV-like	0.210	3.419	1.319	38.7
ThinKV-like	0.210	3.583	1.482	44.6
LongFlow-like	0.210	3.961	1.861	74.2
ThoughtFlow	0.210	3.961	1.861	74.2
ThoughtFlow-recent	0.210	3.562	1.461	44.2
ThoughtFlow-saliency-recent	0.210	3.434	1.333	38.1
Held-out R-KV-like	0.211	3.482	–	43.7
Held-out sweep best	0.211	3.480	–	43.7

Table 1: Retained-context continuation proxy on distilgpt2 traces. Lower NLL is better. Full context is not a matched-budget baseline. The held-out sweep row is selected on 12 train traces and reported on 12 held-out traces, so its NLL is not directly averaged with the all-trace rows.

2 Hidden/KV Telemetry

We then ran a Mac-local telemetry gate using real distilgpt2 tensors rather than only text markers. The probe ranks retained tokens by attention received, final-hidden norm, key norm, value norm, or combined KV norm. This is still not sparse-KV decoding, but it checks whether the interpretable phase policy is merely preserving labels that hidden/KV saliency would not select.

Policy	Keep rate	Anchor recall	Phase recall	Math-state recall
ThoughtFlow	0.207	1.000	1.000	0.918
LongFlow-like	0.207	1.000	1.000	0.918
ThinKV-like	0.207	1.000	0.948	0.848
`value_norm_topk`	0.207	0.000	0.492	0.845
train-selected ThoughtFlow sweep	0.207	1.000	0.430	0.956

Table 2: Hidden/KV saliency telemetry on 24 distilgpt2 traces. The strongest real-saliency proxy by phase+math recall is `value_norm_topk`. ThoughtFlow beats it on phase recall, but LongFlow-like ties ThoughtFlow under the local importance heuristic.

Paired against `value_norm_topk`, ThoughtFlow's phase-recall delta is +0.508 with 95% CI [+0.436, +0.579]. The math-state delta is only +0.073 with 95% CI [-0.078, +0.223]. This rules out a reviewer-facing claim based on phase recall alone: it may be marker preservation rather than useful cache retention.

3 Decision

The branch is mixed but not revived as a positive method. ThoughtFlow-saliency-recent repairs most of the phase-only failure, and a small held-out sweep reaches a tie-range result against R-KV-like at matched budget. The hidden/KV telemetry is useful for diagnosis, but it does not provide a quality result and does not separate ThoughtFlow from the local LongFlow-like importance proxy. A paper-ready claim would require a train-fixed ThoughtFlow successor to beat R-KV-like, LongFlow-like, and ThinKV-like rows at matched bytes under actual sparse-KV or cache-dropping execution.

4 Artifacts

- `experimental/thoughtflow_fp8/phase2/perplexity_impact_proxy.md`
- `experimental/thoughtflow_fp8/phase2/policy_sweep.md`
- `experimental/thoughtflow_fp8/phase2/hidden_saliency_retention_probe.md`
- `experimental/thoughtflow_fp8/phase2/real_trace_retention_sweep.md`
- `experimental/thoughtflow_fp8/paper/colm_workshop_shell.md`